

Clustering Laps from Weather

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CSE 4224 – Introduction to Machine Learning

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Project Goals

Goals

- Predict surface conditions (Clear, Rain, Snow, Tornado) based on lap telemetry
- Understand which driving features help separate weather types
- Compare clustering effectiveness
 - Custom K-Means vs. Scikit-Learn
 - K-Means vs. Other Clustering Methods
- Learn how to improve clustering accuracy

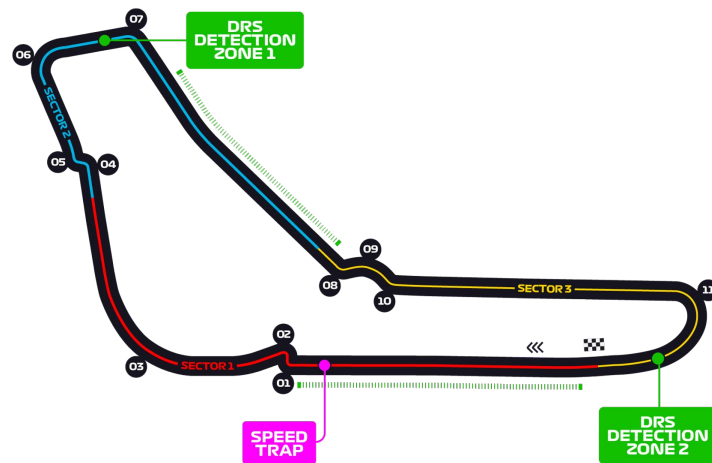




Project Data

Data Collection

- Game: Assetto Corsa
- Track: Monza
- Car: 2017 Porsche 911 Cup Car
- Logged telemetry data using SRT
- Features consist of speed, throttle, brakes, suspension, etc



Data Preparation

- Removed laps labeled 0 and 9 (laps 1 and 10) due to lack of speed and crashes
- Aggregated laps by lap time (maximum time for each respective lap) and mean throttle, brake, and rpm.
- Converted to numerical data
- Resulted in 32 usable laps (8 laps per weather condition)

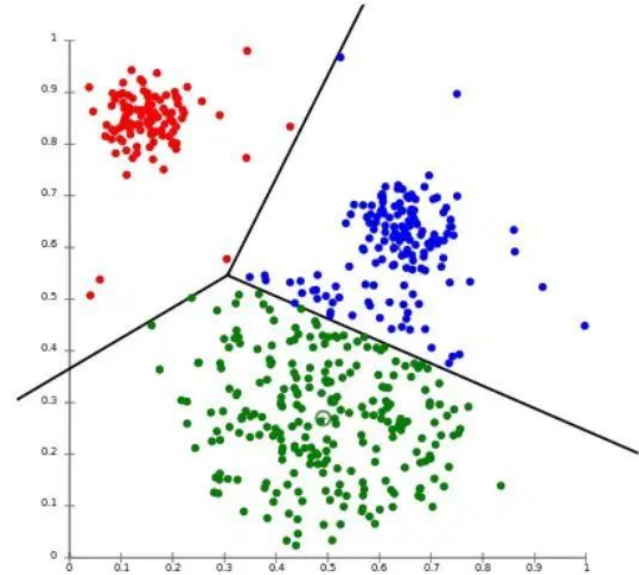
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=== Aggregated lap data ===
Weather lap_number lap_time throttle_mean brake_mean rpm_mean
Clear 1 111.822 0.845107 0.098032 7410.074753
Clear 2 111.657 0.844229 0.098351 7408.760549
Clear 3 111.723 0.843273 0.098148 7402.717572
Clear 4 111.711 0.841865 0.097926 7403.245355
Clear 5 111.606 0.841606 0.097906 7406.018493
Clear 6 111.591 0.842803 0.097579 7404.677375
Clear 7 111.624 0.841646 0.097789 7406.837559
Clear 8 111.528 0.842958 0.097801 7408.204810
Rain 1 113.793 0.838232 0.100901 7371.102796
Rain 2 113.766 0.837741 0.101436 7369.526567
Rain 3 113.748 0.836871 0.100372 7372.538982
Rain 4 113.838 0.836695 0.100729 7368.545320
Rain 5 113.835 0.836740 0.100304 7370.802309
Rain 6 113.682 0.838278 0.100586 7370.255079
Rain 7 113.628 0.835202 0.100709 7375.594548
Rain 8 113.487 0.836527 0.100875 7380.457371
Snow 1 113.532 0.837439 0.100431 7377.206894
Snow 2 113.448 0.837277 0.100385 7379.849106
Snow 3 113.310 0.838326 0.101013 7379.323407
Snow 4 113.364 0.837930 0.100693 7373.350061
Snow 5 113.196 0.838016 0.100596 7377.351276
Snow 6 113.208 0.837540 0.100473 7379.050356
Snow 7 113.151 0.837364 0.100168 7379.435753
Snow 8 113.190 0.836930 0.100275 7376.224779
Tornado 1 113.916 0.835832 0.101062 7366.475690
Tornado 2 113.901 0.833808 0.100915 7364.430978
Tornado 3 113.874 0.836952 0.100740 7366.860566
Tornado 4 113.904 0.835984 0.100313 7363.278955
Tornado 5 113.844 0.836569 0.100544 7364.102622
Tornado 6 113.793 0.835076 0.100464 7367.891040
Tornado 7 113.751 0.834645 0.100355 7368.878451
Tornado 8 113.781 0.835218 0.100202 7366.528651
```



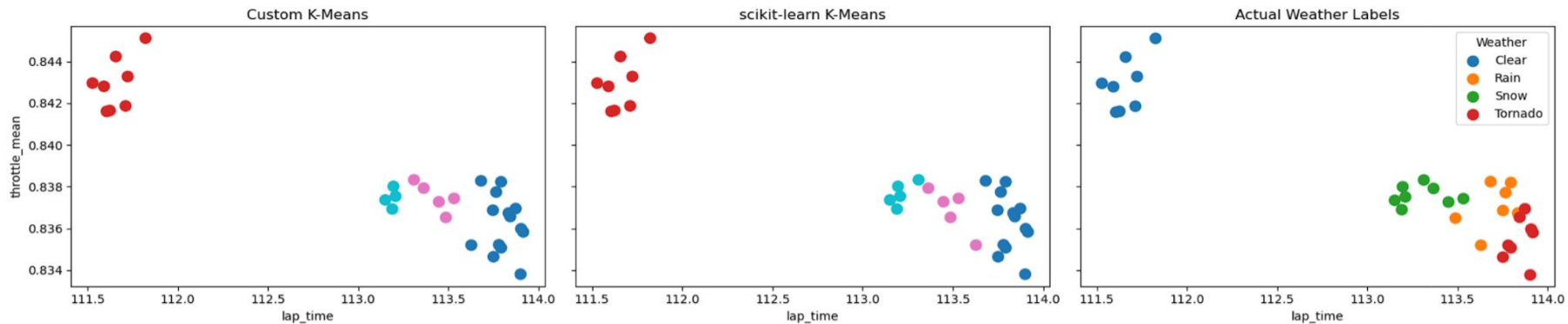
K-Means

K-Means Preparation

- Built from scratch using the Euclidean distance
- Random centroid initialization (seeded for consistency)
- Iterative assignment and centroid update
- Silhouette tuning and purity comparison



K-Mean Results



- 'Clear' and 'Tornado' clustered cleanly
- 'Rain' and 'Snow' had overlapping behaviors

K-Mean Results

=== Confusion Matrix for Custom K-Means ===

Custom Cluster 0 1 2 3

Weather

Clear 0 8 0 0

Rain 7 0 1 0

Snow 0 0 4 4

Tornado 8 0 0 0

Purity (Custom) = 0.75

=== Confusion Matrix for scikit-learn K-Means ===

sklearn Cluster 0 1 2 3

Weather

Clear 0 8 0 0

Rain 6 0 2 0

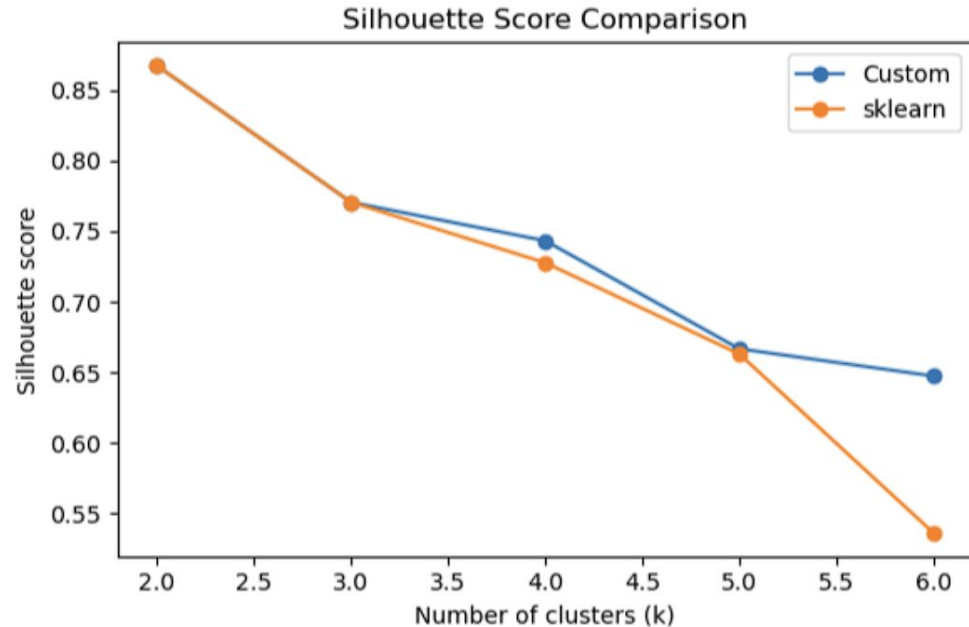
Snow 0 0 3 5

Tornado 8 0 0 0

Purity (sklearn) = 0.75

=== Silhouette Scores by Cluster Count (k) ===

k	silhouette_custom	silhouette_sklearn
2	0.867256	0.867256
3	0.770560	0.770560
4	0.743318	0.727733
5	0.666734	0.663104
6	0.647454	0.536413



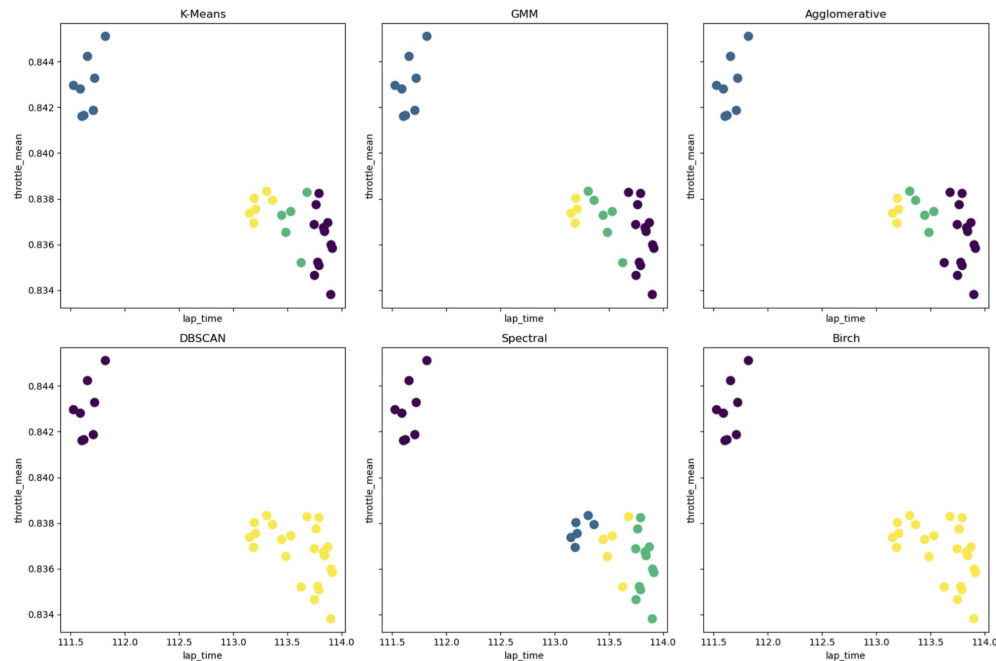
➤ 2 clusters = best separation, 4 clusters = best meaning



Alternative Methods

Alternative Method Preparation

- Gaussian Mixture Model (GMM)
 - Agglomerative Clustering
 - DBSCAN
 - Spectral Clustering
 - Birch
- GMM and Agglomerative = better for subtle differences
- DBSCAN classified outliers as noise
- Spectral is similar to K-Means





Conclusion

Conclusions

- Custom KMeans is comparable to sklearn when configured well
- Clustering can reveal weather surface types
- Though perfect separation wasn't achieved, the clusters revealed real-world patterns, especially between dry and non-dry conditions
- Could always use a bigger dataset!





Questions?